

Photoshopping the Selfie: Self Photo Editing and Photo Investment are Associated with Body Dissatisfaction in Adolescent Girls

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ABSTRACT

Objective: Social media engagement by adolescent girls is high. Despite its appeal, there are potential negative consequences for body dissatisfaction and disordered eating from social media use. This study aimed to examine, in a cross-sectional design, the relationship between social media use in general, and social media activities related to taking “selfies” and sharing specifically, with overvaluation of shape and weight, body dissatisfaction, and dietary restraint.

Method: Participants were 101 grade seven girls ($M_{\text{age}} = 13.1$, $SD = 0.3$), who completed self-report questionnaires of social media use and body-related and eating concerns measures.

Results: Results showed that girls who regularly shared self-images on social media, relative to those who did not, reported significantly higher overvaluation of shape and weight, body dissatisfaction, dietary restraint, and

internalization of the thin ideal. In addition, among girls who shared photos of themselves on social media, higher engagement in manipulation of and investment in these photos, but not higher media exposure, were associated with greater body-related and eating concerns, including after accounting for media use and internalization of the thin ideal.

Discussion: Although cross-sectional, these findings suggest the importance of social media activities for body-related and eating concerns as well as potential avenues for targeted social-media-based intervention. © 2015 Wiley Periodicals, Inc.

Keywords: social media; self-photos; overvaluation of shape and weight; dietary restraint; body dissatisfaction; adolescent girls

(*Int J Eat Disord* 2015; 48:1132–1140).

Introduction

Sociocultural theories of body dissatisfaction and dietary restraint propose that pressures from media, family, and peers contribute to developing and maintaining body dissatisfaction.¹ Some scholars worry that media may be one source of poten-

tial influence. Although evidence has been mixed, some experimental^{2,3} and cross-sectional^{4,5} studies suggest that exposure to thin-ideal media is associated with body dissatisfaction, findings also supported by some,⁶ but not all^{7,8} meta-analyses. Some prospective studies indicate that thin-ideal media exposure⁹ and perceived media pressure to be thin¹⁰ predict body dissatisfaction increases, although these findings are not consistently observed.^{11,12}

Researchers have begun to examine new media, including Internet and online activities such as social media engagement including Facebook, Instagram, Snapchat, and Pinterest. An early study showed that higher appearance-related Internet use, but not television-based exposure to appearance-related media, was associated with lower weight satisfaction and higher drive for thinness in 15-year-old girls.¹³ Adolescent¹⁴ and pre-adolescent¹⁵ girls' exposure to the Internet (excluding homework) has also correlated with internalization of appearance ideals and body surveillance.

Accepted 2 August 2015

Additional Supporting Information may be found in the online version of this article.

Supported by a grant from the La Trobe University Building Healthy Communities Research Focus Area awarded to Susan J Paxton, Eleanor H Wertheim, and Jennifer Masters.

Disclosures of Conflict of Interest: The authors have no conflicts to report.

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Published online 27 August 2015 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/eat.22449

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While these findings point to associations between Internet use and body-related concerns, some types of Internet use may be particularly problematic. Researchers have, therefore, begun investigating nuanced aspects of Internet usage, specifically social media, used by 81% of adolescent Internet viewers.¹⁶ Social media use may have stronger associations with body dissatisfaction than other Internet-related activities¹⁵ because it is highly interactive and self-exposing, with others commenting on postings in a public manner.¹⁷ Furthermore, posting self-images exposes females to greater peer scrutiny of appearance and competition, which, from an evolutionary perspective, has been proposed to increase body dissatisfaction and body concerns.¹⁸

Findings relating Internet and social media use to body image and dietary restraint have been inconsistent. Whereas one study of adolescent females found greater self-reported social media use was associated with body-related concerns and drive for thinness,¹⁴ another study found no such associations.¹² In two studies time spent by adult women,^{19–21} but not men²¹ on social media and/or communication websites, correlated with body-related concerns and drive for thinness. However, a further college student study found more time spent on Facebook was associated with less, not more, appearance investment.²²

More in-depth investigation of specific social media activities may clarify existing inconsistent findings.²³ Specifically, posting photos of oneself is common practice, with 91% of teens who use social media posting photos of themselves.¹⁶ Taking “selfies” (photos one takes of oneself) or “usies” (which include others), and sharing and comparing these through social media, may heighten appearance focus and increase internalization of appearance ideals, contributing to body dissatisfaction. A study examining specific Facebook applications in adolescent girls supports this view.²⁴ Engagement in *photo-based* Facebook activities was correlated with greater thin ideal internalization, self-objectification, and drive for thinness, whereas *overall* Facebook and internet use were unrelated to body image.²⁴

This study aimed to examine relationships between social media activities, including sharing self-photos, and body dissatisfaction, overvaluation of shape and weight, and dietary restraint in early adolescent girls. Early adolescents were selected as this is a period that has been identified as appropriate for prevention³ and improving our understanding of relationships with modifiable

social media activities could inform intervention program development. Social media use, particularly interactions that heighten appearance focus, are likely to be associated with body-related and eating concerns, moreso than general digital media use.

The second aim was to understand how photo investment and photo manipulation aspects of sharing self-photos are associated with body-related and eating concerns. To date these aspects of self-photo sharing have not been examined. It may not be sharing self-photos that is problematic for body-related and eating concerns, but the approach to doing so. An attitudinal construct we are proposing is photo *investment*, which reflects a person's concerns about photo quality, how photos portray the individual, and effort expended in choosing self-photos before associated sharing. An associated construct, photo *manipulation* (commonly available using editing programs) involves altering the appearance of photo elements, such as people's features, prior to sharing.

Theoretical models of eating disorders conceptualize appearance-focused behaviors, such as body checking and avoidance, as manifestations of and *maintaining factors* for body dissatisfaction and overvaluation of shape and weight, core aspects of eating disorders pathology.²⁵ Photo-related behaviors such as photo manipulation could be conceptualized similarly, with such behaviors likely to be undertaken by those with greater body dissatisfaction, due to higher investment in appearance,²⁶ which maintains persistent body concerns. Similar to body checking, self-photo-taking and sharing on social media focus attention on appearance, possibly contributing to preoccupation in those with greater body-related and eating concerns.

In summary, we hypothesized that girls who engaged frequently in social media photo activities, selfie-taking and sharing, would report greater overvaluation of shape and weight, body dissatisfaction, and dietary restraint. We also hypothesized that, among girls who share self-photos on social media, those who reported greater photo investment and manipulating photos before sharing would report greater overvaluation of shape and weight, body dissatisfaction, and dietary restraint, including after accounting for known risk factors, media exposure⁹ and internalization of the thin ideal.²⁷ Findings will assist theory development and targeting activities that may benefit from inclusion in body-related and eating concern interventions.

Method

Sample

Participants were 101 female grade seven students with a mean age of 13.13 years ($SD = 0.33$). Most participants (91%) were born in Australia, with others (9%) born in East Asia, Europe, and New Zealand. Participants' parents were mostly Australian born (73% mothers; 69% fathers) with others born in East Asia (8%; 5%), Europe (9%; 8%), other countries (6%; 12%); or countries unknown to participants (4%; 6%). The sample was recruited from one co-educational public and one single-sex independent secondary school in metropolitan Melbourne which were participating in a body dissatisfaction intervention. Data from the baseline assessment are reported here.

Measures

Demographic Information. Participants provided information regarding their age, country of birth, and their parents' countries of birth, if known.

Media Exposure. Hours per day of digital media exposure were assessed with a 6-item version of a media use questionnaire.²⁸ Adequate test-retest reliability and construct validity have been demonstrated in adolescent samples.²⁸ For this study, the magazine exposure item was replaced with an item assessing smart phone/tablet use to reflect contemporary digital media use. Participants indicated the amount of time (0 min to 5+ h per day excluding school work and homework) on weekdays and weekend days they use TV/DVDs, computers, and phone/tablets. Items were weighted according to weekday and weekend day use and summed for a total score indicating average number of hours per day exposed to digital media. Scores range from 0 to 15 h. Cronbach's alpha was 0.72.

Social media and online engagement was assessed with the social media and digital communications scale developed by the authors for this research. The scale assessed generic social media uses, rather than specific sites, so that rapid changes in social media preferences would not render the measure obsolete. Participants' responses to eight items were recorded as 1 (*yes*) or 0 (*no*) to indicate which forms of communication they used, including text messages, email, Twitter, social networking, instant messaging, virtual worlds, online video sharing, and online photo sharing. Responses were summed to form a total score for social media engagement with higher scores indicating use of more forms of social and online media. Cronbach's alpha was not calculated as it is inappropriate for an index measure.²⁹

Photo Activities. A measure was developed to assess practices of taking and sharing digital images online. The first author developed an item pool, which all authors

reviewed for face validity and clarity. Items were revised and omitted by consensus.

Selfe (self-image) taking frequency. Frequency of taking self-images was assessed with two items that asked how frequently participants take "selfies" with only themselves in the photo and "usies" with themselves and others in the photo, rated from 1 (*less than once a month*) to 8 (*more than twice a day*). The mean of the two items was calculated with higher scores indicating higher frequency of taking photos of self. Spearman-Brown (internal consistency) coefficient for two-item scales³⁰ was 0.86.

Selfe (self-image) sharing. The frequency with which participants share photos of self through social media was assessed with two items: "Do you post photos of yourself online or share them through services like 'Snapchat' or 'Instagram'?" and "Do you avoid putting photos of yourself on social media?" (reverse scored) rated from 1 (*never*) to 5 (*always*). The mean of items was calculated to form a scale score with higher scores reflecting more frequent photo-posting activity. Spearman-Brown's coefficient was 0.82.

Participants who shared photos of themselves online at least sometimes (3, 4, or 5 on the response scale) were considered regular self-photo sharers and were asked further questions about their photo sharing activities.

Photo investment. Eight items assessed investment and effort participants expend choosing photos of themselves to share on social media and concern they have about such posts. Items were presented along a visual analogue scale from 0 to 100 and were anchored by opposing statements such as "It's easy to choose the photo" and "It's hard to choose the photo". The mean for items was calculated for the scale score with higher scores reflecting higher investment in photo sharing through social media. Cronbach's alpha was 0.85.

Photo manipulation. Participants responded to 10 items from 1 (*never*) to 5 (*always*) to indicate the extent to which they manipulated or edited photos of themselves prior to sharing. Example items were "Edit or use apps to smooth skin" and "Make yourself look skinnier". Total scores, ranging from 10 to 50, were calculated from the sum of item responses and higher scores reflected more frequent photo manipulation. Cronbach's alpha was 0.85. Items from the photo investment and photo manipulation scales are shown in Supporting Information Appendices A and B, respectively.

To establish test-retest reliability of the self-photo activity measures, intraclass correlations coefficients (ICC) for agreement (two-way random effects models) were computed for test (T1) and retest (T2) assessments conducted 4 weeks apart with a subsample of participants from this study ($n = 30$; $M_{Age} = 13.10$, $SD = 0.37$).

Excellent agreement was indicated for self-photo frequency (ICC = 0.91, $p < 0.001$; $M_{T1} = 3.98$, $SD_{T1} = 2.08$; $M_{T2} = 3.75$, $SD_{T2} = 2.08$), self-photo sharing (ICC = 0.96, $p < 0.001$; $M_{T1} = 3.08$, $SD_{T1} = 1.16$; $M_{T2} = 3.10$, $SD_{T2} = 1.34$) and self-photo investment (ICC = 0.93, $p < 0.001$; $M_{T1} = 52.92$, $SD_{T1} = 25.02$; $M_{T2} = 51.14$, $SD_{T2} = 22.98$) according to criterion for adequate test-retest reliability (coefficient ≥ 0.70).³¹ Good agreement was indicated for self-photo manipulation (ICC = 0.74, $p < 0.003$; $M_{T1} = 18.62$, $SD_{T1} = 4.92$; $M_{T2} = 18.15$, $SD_{T2} = 6.48$).

Body Image

Body dissatisfaction was assessed with the 10-item Body Dissatisfaction subscale of the Eating Disorders Inventory-3.³² Items such as “I feel satisfied with the shape of my body” were rated on a 6-point scale from 0 (*never*) to 5 (*always*). This measure has adequate test-retest reliability in mixed-age samples³³ and discriminant and convergent validity in nonclinical samples.³² A total score was calculated by summing items with higher scores reflecting higher body dissatisfaction. Cronbach's was 0.92.

Consistent with Grilo and colleagues³⁴ overvaluation of shape and weight was assessed with two items from the Eating Disorder Examination Questionnaire,³⁵ including the item “Over the past 28 days, has your body shape influenced how you think about (judge) yourself as a person”. Items are rated from 0 (*not at all*) to 6 (*markedly*). An overvaluation score is calculated from the mean of the two items with higher scores reflecting higher overvaluation of shape and weight. The Spearman-Brown coefficient was 0.89.

Dietary Restraint

Dietary restraint was assessed with the 10-item Dutch Eating Behavior Questionnaire Restraint subscale.³⁶ Responses to items such as “Do you deliberately eat less in order not to become heavier?” are rated from 1 (*never*) to 5 (*very often*). A total score was calculated from the mean of item responses. Higher scores reflect higher dietary restraint. Scale scores have demonstrated adequate convergent validity in college-aged female samples³⁷ and good test-retest reliability in female adolescent samples.³⁸ Cronbach's alpha was 0.94.

Internalization of the Thin Ideal

The Sociocultural Attitudes Towards Appearance Questionnaire-4—Internalization: Thin/Low Body Fat subscale³⁹ assessed internalization of the thin ideal. Participants responded to five items from 1 (*definitely disagree*) to 5 (*definitely agree*). An example item is “I want my body to look very thin”. The sum of item responses was calculated (range 5–25) with higher scores reflecting higher internalization of the thin ideal. Scores from this recently updated measure have shown stable scale struc-

ture and good convergent validity in US and international college-age female samples.³⁹ Test-retest reliability in the current subsample of participants for whom two assessments were available was adequate (ICC = 0.87, $p < 0.001$; $M_{T1} = 14.17$, $SD_{T1} = 5.60$; $M_{T2} = 14.70$, $SD_{T2} = 5.00$). Cronbach's alpha was 0.84.

Procedure

Ethics approvals were received from the university Human Ethics Committee and state Department of Education and Early Childhood Development. Grade 7 students from participating schools were invited to participate. Written parental consent and participant assent was provided for and by participants. The consent rate was 47.3%. Questionnaire completion was conducted using online survey software during class time and was supervised by a researcher and class teacher.

Data Analysis

Participants were categorized as self-photo sharers if they reported sharing photos sometimes, often, or always. Nonsharers reported sharing images of self either never or rarely. Descriptive statistics and correlations between variables were calculated for the total sample, and separately in the photo-sharer subsample (and in the nonsharer subsample for descriptive statistics only).

To examine differences in body-related and eating concerns variables between the sub-groups of photo-sharers and nonsharers, independent samples *t*-tests were calculated. To examine the amount of variance in overvaluation of shape and weight, body dissatisfaction, and dietary restraint accounted for by independent variables, separate hierarchical linear regressions were conducted with the subsample of participants who were classified as photo sharers. Due to the high correlation between self-image frequency and self-image sharing ($r = 0.68$), they were combined using factor scores from principal components analysis into a single variable representing self-image activity, which was used as a predictor in the analyses. Multicollinearity was not indicated for other independent variables with the highest correlation between variables being $r = 0.50$, the highest VIF value 1.47, and lowest tolerance value 0.68. Media exposure and social media engagement were entered in step 1. These were followed by internalization of the thin ideal (step 2), self-image activities (step 3), and self-photo investment and self-photo manipulation (step 4).

TABLE 1. Means and standard deviations for media exposure, self photo-posting variables, and body-related and eating concerns and independent samples *t*-tests comparisons between photo-sharers and non-photo-sharers

	Mean (<i>SD</i>)			<i>t</i>	η^2
	Total Sample (<i>N</i> = 101)	Photo-Sharers (<i>n</i> = 74)	Non-Photo-Sharers (<i>n</i> = 27)		
Overvaluation shape and weight	2.37 (2.04)	2.61 (2.06)	1.70 (1.85)	−2.02 ^a	0.04
Body dissatisfaction	22.69 (12.20)	24.22 (11.65)	18.52 (12.90)	−2.11 ^a	0.04
Dietary restraint	2.27 (1.00)	2.38 (1.02)	1.97 (0.93)	−1.81	0.03
Internalization of the thin ideal	15.09 (5.04)	15.93 (4.84)	12.78 (4.96)	−2.88 ^b	0.08
Digital media exposure	4.74 (2.78)	5.19 (2.84)	3.53 (2.25)		
Social media engagement	5.00 (1.62)	5.50 (1.28)	3.63 (1.67)		
Self-image frequency	4.03 (2.19)	4.86 (1.91)	1.74 (0.89)		
Self-image posting	3.25 (1.26)	3.84 (0.83)	1.65 (0.72)		
Self-photo manipulation		20.16 (6.99)			
Self-photo investment		57.59 (20.09)			

^a*p* < 0.05.^b*p* < 0.01. η^2 0.01 = small; 0.06 = moderate; 0.14 = large.**TABLE 2.** Correlations between media exposure and self-photo-posting variables and body-related and eating concerns for the full sample (above the diagonal) and self-photo-sharers (below the diagonal)

	1	2	3	4	5	6	7	8	9
Overvaluation shape and weight	—	0.77 ^c	0.69 ^c	0.61 ^c	0.28 ^b	0.10	0.17		
Body dissatisfaction	0.78 ^c	—	0.67 ^c	0.74 ^c	0.21 ^a	0.14	0.15	NA	NA
Dietary restraint	0.69 ^c	0.75 ^c	—	0.64 ^c	0.15	0.15	0.18 ^d	NA	NA
Internalization of the thin ideal	0.56 ^c	0.72 ^c	0.66 ^c	—	0.23 ^a	0.16 ^d	0.24 ^a	NA	NA
Digital media exposure	0.21 ^d	0.09	0.12	0.12	—	0.39 ^c	0.33 ^b	NA	NA
Social media engagement	0.03	0.07	0.17	0.06	0.22 ^d	—	0.51 ^c	NA	NA
Self-photo activities	0.03	0.07	0.11	0.03	0.22 ^a	0.22 ^a	—	NA	NA
Self-photo investment	0.60 ^c	0.67 ^c	0.52 ^c	0.55 ^c	−0.01	0.08	0.08	—	NA
Self-photo manipulation	0.37 ^b	0.40 ^b	0.44 ^c	0.36 ^b	−0.02	−0.03	−0.04	0.42 ^c	—

NB: values above the diagonal are for the complete sample (*N* = 101); values below the diagonal are for the subsample of regular self-photo-sharers (*n* = 73).^a*p* < 0.05.^b*p* < 0.01.^c*p* < 0.001.^d*p* < 0.10.

Results

Participant Characteristics

Means and standard deviations for variables for the whole sample and for the subsamples of self-photo sharers and non-sharers are presented in **Table 1**. On average, participants reported being exposed to a little under 5 h per day of digital media and used five different types of social media. The most frequently used forms of social media were text messages (98%), emails (95%), and social networks (86%). The least frequently used forms were Twitter (20%) and virtual worlds (20%). Participants were regular, but not prolific self-photo takers with 50.5% of participants taking “selfies” at least once per fortnight (2 week period) and 49.5% of participants taking “usies” at least once per week. Among participants, 73.3% were categorized as regular self-photo sharers, sharing photos sometimes, often, or always, and on average, participants reported rarely engaging in manipulation of self-photos. A moderate proportion of the sample

(*n* = 30; 29.7%) scored above the adult cut-off of ≥ 5 for clinically significant levels of overvaluation of shape and weight.³⁴

Independent samples *t*-tests showed that self-photo sharers had significantly higher mean scores for overvaluation of shape and weight, body dissatisfaction, and internalization of the thin ideal than non-sharers. Effect sizes were generally small. The difference between groups for dietary restraint failed to reach significance (*p* = 0.07).

Relationships between Social Media and Self Photo Activities and Body-Related and Eating Concerns

Table 2 shows the correlations between variables for the total sample and the subsample of self-photo sharing participants. Overvaluation of shape and weight, body dissatisfaction, dietary restraint, and internalization of the thin ideal were highly intercorrelated. Media exposure, social media engagement, and self-image variables had small

TABLE 3. Summary statistics for hierarchical linear regression analyses predicting overvaluation of shape and weight, body dissatisfaction, and dietary restraint in self-photo-sharers ($N = 73$)

	R^2	ΔR^2	$F(df)$	β
<i>DV: Overvaluation of shape and weight</i>				
Step 1	0.04	0.04	1.60 (2.70)	
Digital media exposure				0.21 ^d
Engagement with social media				−0.02
Step 2	0.34	0.30 ^c	11.77 (3.69) ^c	
Internalization of the thin ideal				0.55 ^c
Step 3	0.39	0.00	8.76 (4.68) ^c	
Self-photo activities				0.04
Step 4	0.49	0.15 ^c	10.61 (6.66) ^c	
Self-photo investment				0.43 ^c
Self-photo manipulation				0.07
<i>DV: Body dissatisfaction</i>				
Step 1	0.01	0.01	0.37 (2.70)	
Digital media exposure				0.08
Engagement with social media				0.05
Step 2	0.39	0.38 ^c	14.67 (3.69) ^c	
Internalization of the thin ideal				0.62 ^c
Step 3	0.39	0.00	10.84 (4.68) ^c	
Self-photo activities				0.00
Step 4	0.57	0.18 ^c	14.38 (6.66) ^c	
Self-photo investment				0.47 ^c
Self-photo manipulation				0.05
<i>DV: Dietary restraint</i>				
Step 1	0.04	0.04	1.28 (2.70)	
Digital media exposure				0.09
Engagement with social media				0.15
Step 2	0.38	0.34 ^c	13.83 (3.69) ^c	
Internalization of the thin ideal				0.59 ^c
Step 3	0.38	0.00	10.41 (4.68) ^c	
Self-photo activities				0.07
Step 4	0.47	0.09 ^b	9.69 (6.66) ^c	
Self-photo investment				0.23 ^a
Self-photo manipulation				0.19 ^d

^a $p < 0.05$.^b $p < 0.01$.^c $p < 0.001$.^d $p < 0.10$.

positive correlations with overvaluation of shape and weight, body dissatisfaction, dietary restraint, and internalization of the thin ideal in the total sample, although not all correlations were statistically significant. Within the self-photo sharing group, high scores for self-photo investment and self-photo manipulation were associated with high scores for body-related and eating concerns.

Hierarchical regression analyses conducted within the subsample of self-photo-sharers revealed that when controlling for digital and social media use (step 1), the addition of internalization of the thin ideal at step 2 made a significant contribution to the explained variance in each of overvaluation of shape and weight, body dissatisfaction, and dietary restraint. The addition of self-image activities in step 3 did not explain additional variance for any of the dependent variables. Photo investment, added in step 4, explained significant additional variance for overvaluation of shape and weight, body dissatisfaction, and dietary restraint. The addition of

photo manipulation, also in step 4, explained variance (at trend level $p = 0.065$) in dietary restraint, but did not explain significant variance in overvaluation of shape and weight or body dissatisfaction. **Table 3** displays the summary statistics.

Discussion

Our study aimed to examine the association between social media self-photo-related activities and overvaluation of shape and weight, body dissatisfaction, and dietary restraint in an early adolescent female sample. Overall, results generally supported our hypotheses whereby higher levels of body-related and eating concerns were found for participants engaging in more social media-related self-photo activities, including online photo sharing, and higher frequency of manipulation of photos for online posting, but concerns were unrelated to participants' level of media exposure. In

addition, for those who shared photos higher investment in self-photo activities was associated with greater overvaluation of shape and weight, and body dissatisfaction.

Our findings revealed that young adolescent girls reported engaging with multiple forms of digital and social media. Approximately half the sample took self-photos at least weekly and almost three quarters were regular self-photo sharers, although engaging in these social media activities was not associated with body concerns and dietary restraint. We did find that adolescent girls who more regularly shared self-photos reported more overvaluation of shape and weight, body dissatisfaction, and thin ideal internalization, with associations with dietary restraint only at trend level. These findings regarding body-related concerns were consistent with previous studies where greater adolescent use of self-photo-based Facebook activities was associated with greater body related concerns²⁴ and importance of receiving comments on posted photos and greater frequency of comparing photos of oneself with friends' photos being associated with higher levels of disordered eating in college women.²⁰

Higher levels of engagement with general and social media were associated with greater body-related and eating concerns. The strongest correlations in the self-photo sharer subgroup were for photo manipulation and photo investment. These correlations were of higher magnitude than those found in previous studies between Facebook photo activities and eating disorder scores, $r < 0.35$,²⁰ and body-related variables including drive for thinness and internalization of the thin ideal, $r < 0.36$.²⁴ In our study, individuals who more frequently manipulated photos of themselves prior to sharing and reported being more invested in their photos, reported greater body-related and eating concerns at the univariate level, with effect sizes ranging from medium to large.

Importantly, in conservative hierarchical regression analyses, higher investment in self-photo sharing was associated with greater overvaluation of shape and weight, body dissatisfaction, and dietary restraint. However, photo manipulation did not account for unique variance in outcome variables in regression analyses after accounting for previously entered variables. Shared variance of photo manipulation with other variables, particularly internalization of the thin ideal, may explain this latter finding. Additionally, most photo-manipulation items were general in nature, (e.g., "Use a filter to change the overall look of the photo") rather than specific to

body size and shape, which may have reduced associations with body-related and eating concerns. Future research examining editing activities directed specifically at manipulating weight and shape may yield stronger associations.

It is noteworthy that investment in self-images accounted for additional variance in the outcome variables beyond that accounted for by the media variables and internalization of the thin ideal. This extends previous research^{20,24} showing only univariate associations, rather than associations that have persisted in multivariate models. Furthermore, the finding that self-photo investment was a unique cross-sectional predictor of overvaluation of shape and weight and dietary restraint is of interest as it suggests a target, self-photo investment behavior, for intervention.

Taken together, findings of relationships between self-photo-related variables and outcome measures suggest three possible separate, but likely overlapping interpretations. First, young adolescent girls with high levels of body-related and eating concerns might engage in social media⁴⁰ activities that are appearance focused such as self-photo manipulation and investment in efforts to present an ideal appearance when sharing images. Second, consistent with Prieler and Choi's²³ conceptualization, the relationship between body-related and eating concerns and social media use found in this study appears to be differentiated by the passive or active nature of the social media platform. In contrast to the small or null relationships for general and social media engagement, the robust findings for the self-photo-related variables suggests that the active component of social media engagement contributes to negative outcomes. Specifically, actively presenting a desired image, or actively seeking, comparing, and commenting on others' images may provoke self-scrutiny that leads to body concerns in a manner that does not occur with passive traditional media engagement. Third, in line with Perloff's¹⁷ proposed model of social media and body image, engagement in social media may be sought by individuals with high body-related and eating concerns who seek gratification for their concerns, such as "reassurance and validation regarding physical and social attractiveness".¹⁷

The interpretations for findings outlined above are consistent with the proposition that active social media use may be a maintaining factor for body-related and eating concerns. In this manner, the relationship between social media and body-related and eating concerns may be bi-directional and mutually reinforcing such that those who have

body-related and eating concerns would be more likely to be drawn to activities that have a strong appearance focus, and in turn, engagement in appearance focused activities such as self-photo-investment and self-photo-manipulation, contributes to the persistence of body-related and eating concerns. Bi-directional relationships between social media and body-related and eating concerns may be more strongly reinforcing than those seen with traditional media due to both the personalized nature of engagement with social media and greater opportunity for peer interactions, such as appearance comparison and appearance conversations which may mediate the social media-body dissatisfaction relationship.¹⁷ Future research is necessary to examine such sequences.

This study has some limitations. The sample is relatively small, especially in relation to self-photo sharers. The cross-sectional nature of the analyses preclude conclusions being drawn about the direction of relationships observed. Further research is required to clarify whether self-photo-related activities are correlates, maintaining factors, or risk factors for body-related and eating concerns. In addition, our research examined relationships in females and future research should examine the role of social media in boys' body-related and eating concerns.

There are also several measurement points to note. Our measure of engagement with social media, in addition to items about social networking, included items about such activities as text messaging and email which may not normally be as widely public as social networking. Further measurement development is required in this field and differentiation between the effects of public or private photo-sharing may be important. Also in relation to measurement issues, demand characteristics may have affected participant responses. Inclusion of distractor items may ameliorate this problem in future studies.

In summary, this was the first study to examine how self-photo sharing activities, self-image manipulation, and self-image investment relate to body-related and eating concerns, and contributes new social media-focused measures with good psychometric properties. The study provides preliminary evidence to suggest a role for social media-related self-photo activities, particularly photo investment, in the maintenance of body-related and eating concerns. The ways in which people use media, through manipulation of and investment in self-images prior to sharing on social media appear to be especially important for these concerns. The

findings highlight the potential value of addressing self-photo-related activities in prevention and early intervention of body dissatisfaction and disordered eating.

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